Lab: Thought Experiment — Perception as Inference

Objective

Investigate the philosophical implications of treating perception as inference through structured thought experiments, phenomenological reflection, and conceptual analysis.

Part 1: The Müller-Lyer Illusion

Goal: Examine what visual illusions reveal about the nature of perception.

The Müller-Lyer illusion causes two lines of equal length to appear different due to arrowhead configurations. Even knowing the lines are equal, the illusion persists.

- What does this persistence tell us about the relationship between perception and belief?
- How would a direct realist (Gibson), a representationalist (Locke), and an inferentialist (Active Inference) each explain this illusion?
- Which explanation do you find most satisfying, and why?

Write your response here

Part 2: The Expert's Eye

Goal: Analyze perceptual learning as evidence for the inferentialist account.

A radiologist can glance at a chest X-ray and immediately see a subtle mass that a novice would miss entirely. Consider:

- Is the radiologist "seeing" something the novice cannot, or "interpreting" what they both see? What hangs on this distinction?

- How does Active Inference account for this difference (hint: refined generative model, higher precision predictions)?
- Does this example support or undermine the claim that perception is "direct"?

Write your response here

Part 3: Controlled Hallucination

Goal: Probe the boundary between perception and hallucination.

Anil Seth calls normal perception "controlled hallucination" — the brain generates predictions that are constrained by sensory input, but hallucinations occur when predictions dominate input.

- Is there a sharp boundary between perception and hallucination, or is it a continuum?
- What are the philosophical implications of calling all perception a form of hallucination? Does this lead to skepticism?
- How would a pragmatist respond to this framing?

Write your response here

Part 4: The Veil of Predictions

Goal: Engage with the epistemological consequences of predictive processing.

If we only ever access our own predictions (updated by prediction error), are we trapped behind a "veil of predictions" — never touching reality itself? This echoes Locke's "veil of ideas" and Kant's distinction between phenomena and noumena.

- Is this a genuine epistemic problem or a pseudo-problem?
- Does the Active Inference framework provide resources to respond to skepticism that classical representationalism does not?
- Write a 100-word response from either the pragmatist or the phenomenological perspective.

Write your response here

Part 5: Synthesis and Position Defense

Goal: Develop your own view on the nature of perception.

Write a 200-word position statement on one of the following:

- "Perception is inference — and this is a deep truth about the nature of mind."
- "The inferentialist account of perception is a useful scientific model but a misleading philosophical theory."
- "Active Inference dissolves the realism/anti-realism debate about perception."

Share with a peer and defend your position against their strongest objection.

Write your response here

Lab Summary

Part	Skill Developed	Key Philosophical Move
1	Theory comparison	Applying multiple philosophical frameworks to a single phenomenon
2	Conceptual analysis	Distinguishing perception from interpretation under inferentialism
3	Boundary reasoning	Exploring the continuum between normal perception and hallucination
4	Epistemological reasoning	Engaging with skeptical implications of predictive processing
5	Position defense	Constructing a view on perception and defending it under criticism